THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 96	Use Brahmagupta's formula.
2	**(a)** 42	Side of square $= 2r$.
3	**(d)** 770	Difference of the two areas.
4	**(b)** 35	Distance $= n \times 2\pi r$.
5	**(c)** 120	Diagonals bisect at right angles: the other diagonal is $2\sqrt{s^2 - (d/2)^2}$.
6	**(d)** 8	The shortest altitude stands on the longest side.
7	**(b)** 484	Side $= \frac{\text{perimeter}}{4}$.
8	**(b)** four times	Reason with the area formulas.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$s = 21$, area $= \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$, using R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$2\pi r = 2 \times \frac{22}{7} \times 7 = 44$, so A is true; R is true but is the area formula, not the reason for A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$s = 23$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)} = 96$ cm^2 (1).	1+1
12	Side of square $= 14$ cm, area $= 196$ cm^2 (1); Circle area $= 154$ cm^2; difference $= 42$ cm^2 (1).	1+1
13	$d_2 = \frac{2\times120}{10} = 24$ cm (1); Side $= \sqrt{5^2 + 12^2} = 13$ cm (1).	1+1
14	$s = 26$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)} = 126$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Radius $= \frac{44}{2\pi} = 7$ cm; Side $= \frac{44}{4} = 11$ cm (1); Circle area $= 154$ cm^2; Square area $= 121$ cm^2 (1); The circle encloses the larger area (1).	1+1+1
16	Boundary $= 2 \times 60 + 2\pi r$; $= 120 + 88 = 208$ m (1); Area $= 60 \times 28 + \pi r^2$ (1); $= 1680 + 616 = 2296$ m^2 (1).	1+1+1
17	$s = 21$, $A = 84$ cm^2 (1); Altitude on 13 cm $= \frac{168}{13} = \frac{168}{13}$ cm (1); Altitude on 14 cm $= \frac{168}{14} = 12$ cm (1).	1+1+1
	OR: Radius $= \frac{88}{2\pi} = 14$ cm; Side $= \frac{88}{4} = 22$ cm (1); Circle area $= 616$ cm^2; Square area $= 484$ cm^2 (1); The circle encloses the larger area (1).	1+1+1
18	Boundary $= 2 \times 80 + 2\pi r$; $= 160 + 44 = 204$ m (1); Area $= 80 \times 14 + \pi r^2$ (1); $= 1120 + 154 = 1274$ m^2 (1).	1+1+1
	OR: $s = 30$, $A = 150$ cm^2 (1); Altitude on 15 cm $= \frac{300}{15} = 20$ cm (1); Altitude on 20 cm $= \frac{300}{20} = 15$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2\pi r = 2 \times \frac{22}{7} \times 14 = 88$ m (1). (ii) $\pi r^2 = \frac{22}{7} \times 196 = 616$ m^2 (1). (iii) Outer radius $= 21$ m, area $= \frac{22}{7} \times 441 = 1386$ m^2 (1). (iv) Path area $= 1386 - 616 = 770$ m^2 (1); cost $= 770 \times 40 =$ Rs 30800 (1).	1+1+1+2
20	Ring area $= \pi\left(28^2 - 21^2\right)$ (1); $= 1078$ cm^2 (1); Difference in circumferences $= 2\pi \times 7$ (1); $= 44$ cm (1); Units stated (1).	1+1+1+1+1
	OR: $AC = \sqrt{5^2 + 12^2} = 13$ cm (1); Area $\triangle ABC = \frac{1}{2} \times 5 \times 12 = 30$ cm^2 (1); $\triangle ACD$ has sides 13, 14, 15 cm: $s = 21$ cm (1); Area $\triangle ACD = 84$ cm^2 (1); Area $ABCD = 114$ cm^2 (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up circumference ($2\pi r$) and area (πr^2), or writing the area in cm instead of cm^2.
- In Heron's formula, using the perimeter instead of the semi-perimeter s in the factors $(s - a)$.
- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\frac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.

- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2464	Difference of the two areas.
2	**(c)** 28	Distance $= n \times 2\pi r$.
3	**(a)** 240	Diagonals bisect at right angles: the other diagonal is $2\sqrt{s^2 - (d/2)^2}$.
4	**(b)** 9	The shortest altitude stands on the longest side.
5	**(b)** 1936	Side $= \frac{\text{perimeter}}{4}$.
6	**(c)** two times	Reason with the area formulas.
7	**(a)** 150	Use Brahmagupta's formula.
8	**(c)** 378	Side of square $= 2r$.
9	**(c)** A is true but R is false.	A is correct; R is false because the product of adjacent sides is not the area unless the angle between them is 90°.
10	**(d)** A is false but R is true.	Area $= \frac{22}{7} \times 49 = 154$ cm^2, so A is false (44 is the circumference); R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Side of square $= 28$ cm, area $= 784$ cm^2 (1); Circle area $= 616$ cm^2; difference $= 168$ cm^2 (1).	1+1
12	$d_2 = \frac{2 \times 96}{16} = 12$ cm (1); Side $= \sqrt{8^2 + 6^2} = 10$ cm (1).	1+1
13	$s = 29$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)} = 150$ cm^2 (1).	1+1
14	Side of square $= 42$ cm, area $= 1764$ cm^2 (1); Circle area $= 1386$ cm^2; difference $= 378$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Radius $= \frac{132}{2\pi} = 21$ cm; Side $= \frac{132}{4} = 33$ cm (1); Circle area $= 1386$ cm^2; Square area $= 1089$ cm^2 (1); The circle encloses the larger area (1).	1+1+1
16	Boundary $= 2 \times 70 + 2\pi r$; $= 140 + 132 = 272$ m (1); Area $= 70 \times 42 + \pi r^2$ (1); $= 2940 + 1386 = 4326$ m^2 (1).	1+1+1
17	$s = 34$, $A = 204$ cm^2 (1); Altitude on 17 cm $= \frac{408}{17} = 24$ cm (1); Altitude on 25 cm $= \frac{408}{25} = \frac{408}{25}$ cm (1).	1+1+1
	OR: Radius $= \frac{176}{2\pi} = 28$ cm; Side $= \frac{176}{4} = 44$ cm (1); Circle area $= 2464$ cm^2; Square area $= 1936$ cm^2 (1); The circle encloses the larger area (1).	1+1+1
18	Boundary $= 2 \times 90 + 2\pi r$; $= 180 + 88 = 268$ m (1); Area $= 90 \times 28 + \pi r^2$ (1); $= 2520 + 616 = 3136$ m^2 (1).	1+1+1
	OR: $s = 35$, $A = 210$ cm^2 (1); Altitude on 20 cm $= \frac{420}{20} = 21$ cm (1); Altitude on 21 cm $= \frac{420}{21} = 20$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $s = \dfrac{13+14+15}{2} = 21$ m (1). (ii) $\sqrt{21 \times 8 \times 7 \times 6} = 84$ m^2 (1). (iii) $\frac{1}{2} \times 14 \times h = 84 \Rightarrow h = 12$ m (1). (iv) Cost $= 84 \times 30 =$ Rs 2520 (1); altitude on CA $= \dfrac{2 \times 84}{15} = 11.2$ m (1).	1+1+1+2
20	$AC = \sqrt{9^2 + 12^2} = 15$ cm (1); Area $\triangle ABC = \frac{1}{2} \times 9 \times 12 = 54$ cm^2 (1); $\triangle ACD$ has sides 15, 20, 25 cm: $s = 30$ cm (1); Area $\triangle ACD = 150$ cm^2 (1); Area $ABCD = 204$ cm^2 (1).	1+1+1+1+1
	OR: Inner rectangle: 60 m $\times$ 40 m (1); Area $= 2400$ m^2 (1); Bed area $= 616$ m^2 (1); Grass $= 1784$ m^2 (1); Cost $=$ ₹7136 (1).	1+1+1+1+1

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- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\dfrac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.

- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 336	Diagonals bisect at right angles: the other diagonal is $2\sqrt{s^2-(d/2)^2}$.
2	**(d)** 12	The shortest altitude stands on the longest side.
3	**(c)** 3025	Side $=\frac{\text{perimeter}}{4}$.
4	**(a)** $2:1$	Reason with the area formulas.
5	**(a)** 192	Use Brahmagupta's formula.
6	**(a)** 1050	Side of square $=2r$.
7	**(c)** 3696	Difference of the two areas.
8	**(d)** 49	Distance $=n\times 2\pi r$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{90}{360}\times 2\times\frac{22}{7}\times 14=22$ cm, using R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$s=14$, area $=\sqrt{8\times 6\times 8\times 6}=48$; both statements are true, but R only says the formula applies, it does not give the value.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d_2=\frac{2\times 216}{24}=18$ cm (1); Side $=\sqrt{12^2+9^2}=15$ cm (1).	1+1
12	$s=30$ cm (1); $A=\sqrt{(s-a)(s-b)(s-c)(s-d)}=192$ cm^2 (1).	1+1
13	Side of square $=56$ cm, area $=3136$ cm^2 (1); Circle area $=2464$ cm^2; difference $=672$ cm^2 (1).	1+1
14	$d_2=\frac{2\times 240}{30}=16$ cm (1); Side $=\sqrt{15^2+8^2}=17$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Radius $=\frac{220}{2\pi}=35$ cm; Side $=\frac{220}{4}=55$ cm (1); Circle area $=3850$ cm^2; Square area $=3025$ cm^2 (1); The circle encloses the larger area (1).	1+1+1
16	Boundary $=2\times 100+2\pi r$; $=200+176=376$ m (1); Area $=100\times 56+\pi r^2$ (1); $=5600+2464=8064$ m^2 (1).	1+1+1
17	$s=27$, $A=90$ cm^2 (1); Altitude on 12 cm $=\frac{180}{12}=15$ cm (1); Altitude on 17 cm $=\frac{180}{17}=\frac{180}{17}$ cm (1).	1+1+1
	OR: Radius $=\frac{264}{2\pi}=42$ cm; Side $=\frac{264}{4}=66$ cm (1); Circle area $=5544$ cm^2; Square area $=4356$ cm^2 (1); The circle encloses the larger area (1).	1+1+1
18	Boundary $=2\times 50+2\pi r$; $=100+220=320$ m (1); Area $=50\times 70+\pi r^2$ (1); $=3500+3850=7350$ m^2 (1).	1+1+1
	OR: $s=32$, $A=168$ cm^2 (1); Altitude on 25 cm $=\frac{336}{25}=\frac{336}{25}$ cm (1); Altitude on 25 cm $=\frac{336}{25}=\frac{336}{25}$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $25+60+52+39=176$ m (1). (ii) $s=88$ m (1). (iii) $63,\ 28,\ 36,\ 49$ (1). (iv) Area $=\sqrt{63\times 28\times 36\times 49}=\sqrt{3111696}=1764$ m^2 (1); cost $=1764\times 10=$ Rs 17640 (1).	1+1+1+2
20	Inner rectangle: 70 m $\times$ 40 m (1); Area $=2800$ m^2 (1); Bed area $=1386$ m^2 (1); Grass $=1414$ m^2 (1); Cost $=$ ₹2828 (1).	1+1+1+1+1
	OR: Ring area $=\pi\left(35^2-28^2\right)$ (1); $=1386$ cm^2 (1); Difference in circumferences $=2\pi\times 7$ (1); $=44$ cm (1); Units stated (1).	1+1+1+1+1

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